

Cut-and-Choose Issuance for Compact Lattice-Based Keyed-Verification Credentials

Anonymous draft

Working draft — security claims are provisional

Abstract

We investigate a cut-and-choose variant of the keyed-verification blind signature of Beullens, Lyubashevsky, Nguyen, and Seiler (BLNS23). The goal is to remove the expensive first-round zero-knowledge proof that a user-supplied lattice commitment contains correctly evaluated hash functions. A user sends two commitments in each of κ independent lanes. After an unpredictable challenge, the user opens one commitment per lane and the signer evaluates only the complementary commitments. Crucially, the signer releases a single aggregate response rather than the lane responses. A direct chosen-input attack must then evade all κ checks to remove every honestly formed mask, which occurs with probability $2^{-\kappa}$. The resulting credential authenticates a vector of hidden messages and is verified with the tuple of lane keys. We give the construction, its correctness calculation, and a conditional security argument. The central unresolved question is a robust “one honest lane” lemma for adversarially correlated lattice inputs. We identify a four-corner linear splicing attack whenever valid lane inputs can be reused across issuance sessions. The current construction attempts to exclude that attack by deriving every ρ from a unique session identifier. We formalize the remaining requirement as majority-fresh aggregate HMLWE and prove reductions to ordinary HMLWE in two important special cases. The fully adaptive, rotating-majority case required by the construction remains a new assumption. We also give a balanced splicing attack costing $2^{\kappa/2}$ classically (or $2^{\kappa/4}$ with generic quantum search), so the construction remains unproven from standard MLWE and needs larger cut-and-choose parameters than the all-malformed analysis suggests.

Security status. The construction is unproven. Section 5 gives a free break if valid lane inputs are reusable and a balanced malformed-copy splice even with session freshness. It must not be implemented as a secure credential scheme without a full aggregate-HMLWE reduction and revised parameters.

1 Introduction

BLNS23 gives a compact keyed-verification blind signature whose issuance protocol contains two non-interactive zero-knowledge proofs. The user’s proof π_1 establishes knowledge of short $\mathbf{r}$, $\mathbf{e}_2$ and a message m satisfying

$$\rho = H_\rho(\mathbf{r}, \mathbf{e}_2), \quad \mathbf{c} = B\mathbf{r} + \mathbf{e}_2 + H_R(m, \rho).$$

This proof prevents a malicious user from choosing $\mathbf{c}$ so as to query the signing secret on a scaled unit vector. Its statement includes hash evaluation and is the principal issuance bottleneck.

We ask whether statistical cut-and-choose can replace π_1 . A naive construction with N commitments, of which $N - 1$ are opened, has soundness error $1/N$ and therefore requires an infeasible $N \geq 2^\lambda$. Our candidate instead uses κ pairs and returns only one aggregate response. If a single honestly generated unopened lane computationally masks all malformed lanes, a useful chosen-input attack must corrupt every unopened lane simultaneously. Parallel repetition then gives error $2^{-\kappa}$. Applying Fiat–Shamir to the full vector of pair commitments places the challenge and openings in the user’s first message. The signer returns one aggregate response, giving a two-message, round-optimal issuance protocol.

The construction changes the message space. A credential authenticates a vector $(m_0, \dots, m_{\kappa-1})$, not one message copied across all lanes. The opened message in every pair is a decoy; the complementary message remains hidden during issuance and is disclosed at presentation. An application may derive a single payload by hashing the ordered vector.

Status. Correctness and the elementary cut-and-choose probability are immediate. A complete proof of blindness and one-more unforgeability is not supplied here. In particular, Section 9 identifies the one-honest-lane lemma, related-input security, selective failure, and concurrency as open obligations.

2 Preliminaries and notation

Let $R_q = \mathbb{Z}_q[X]/(X^d + 1)$. Vectors are written in bold and all ring operations are modulo q . Let D_s, D_r, D_e be bounded short distributions, and let $\lceil \cdot \rceil_p$ retain the prescribed high-order coefficient bits. The security parameter and number of cut-and-choose lanes are denoted λ and κ . The balanced splicing attack of Section 5 costs only about $2^{\kappa/2}$ classically and $2^{\kappa/4}$ under generic quantum search. Consequently, 128-bit concrete security requires at least $\kappa = 256$ in the classical ROM and $\kappa = 512$ in the QROM before accounting for multi-target and reduction losses.

For lane i , the signer samples

$$\mathbf{s}_i \leftarrow \$ D_s, \quad \mathbf{e}_{1,i} \leftarrow \$ D_e,$$

publishes a random matrix B_i and

$$\mathbf{t}_i^T = \mathbf{s}_i^T B_i + \mathbf{e}_{1,i}^T.$$

The keyed verification key is the tuple $\mathbf{sk} = (\mathbf{s}_0, \dots, \mathbf{s}_{\kappa-1})$. It is generally *not* sufficient to retain only $\sum_i \mathbf{s}_i$, because different lanes authenticate different hash inputs.

We require a batched lattice NIZK Π_{resp} and a 384-bit collision-resistant hash H_{com} . The latter gives approximately 128-bit collision resistance against generic quantum collision search. The NIZK contains no hash preimage relation.

3 Construction

3.1 Commit phase

The user independently constructs two branches $b \in \{0, 1\}$ in each lane $i \in \{0, \dots, \kappa - 1\}$. It samples a 32-byte seed $z_{i,b}$ and uses a domain-separated XOF and canonical deterministic sampler to derive

$$(m_{i,b}, \mathbf{r}_{i,b}, \mathbf{e}_{2,i,b}) = G(\text{sid}, i, b, z_{i,b}),$$

where $m_{i,b}$ is 32 bytes and the two vectors have the prescribed bounded distributions. It computes

$$\begin{aligned} \rho_{i,b} &= H_\rho(\text{sid}, i, b, \mathbf{r}_{i,b}, \mathbf{e}_{2,i,b}), \\ \mathbf{u}_{i,b} &= H_R(\text{credential-lane}, i, m_{i,b}, \rho_{i,b}), \\ \mathbf{c}_{i,b} &= B_i \mathbf{r}_{i,b} + \mathbf{e}_{2,i,b} + \mathbf{u}_{i,b}. \end{aligned}$$

The session identifier sid includes the protocol version, issuer, key epoch, and fresh contributions from both parties. The user commits once to the complete canonically ordered vector of branches:

$$C = H_{\text{com}}(\text{protocol}, \text{issuer}, \text{pk}, \text{sid}, \text{Encode}(\mathbf{c}_{0,0}, \mathbf{c}_{0,1}, \dots, \mathbf{c}_{\kappa-1,0}, \mathbf{c}_{\kappa-1,1})).$$

The encoding includes the branch count, dimensions, and unambiguous lengths. Thus one 48-byte digest binds all 2κ branches and their ordering before challenge selection.

3.2 Fiat–Shamir challenge and opening

After all commitments are fixed, the user derives

$$\chi = (\chi_0, \dots, \chi_{\kappa-1}) = H_{\text{chal}}(\text{protocol}, \text{issuer}, \text{pk}, \text{sid}, C) \in \{0, 1\}^\kappa.$$

For every lane, the user opens branch χ_i by disclosing only the seed z_{i,χ_i} , and sends the full vector $\mathbf{c}_{i,1-\chi_i}$ for the complementary branch. On receipt, the signer first recomputes χ from the transmitted C and verifies that each seed and full vector has the branch role prescribed by χ_i . It expands every disclosed seed, recomputes the opened $\mathbf{c}_{i,\chi_i}$, and combines these with the received complementary vectors in canonical order. It recomputes the global commitment C over all 512 reconstructed branches. It also checks both hash computations, the linear relation, all norm bounds, canonical encodings, and domain

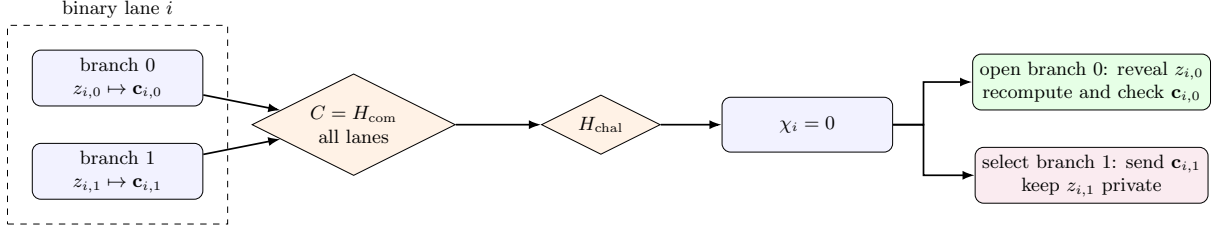


Figure 1: Binary cut-and-choose in one lane, illustrated for $\chi_i = 0$. The challenged branch is regenerated from its seed and checked; only the complementary branch enters the aggregate response. The roles reverse when $\chi_i = 1$.

separators for every opened branch. It aborts without producing any secret-dependent output if any check fails. Define the selected branch $b_i^* = 1 - \chi_i$ and abbreviate $\mathbf{c}_i^* = \mathbf{c}_{i,b_i^*}$.

The selected seeds are retained privately by the user. At presentation the user reveals only (m_i^*, ρ_i^*) , never z_{i,b_i^*} : revealing a selected seed would let the issuer reconstruct $\mathbf{r}_i^*, \mathbf{e}_{2,i}^*$ and link the presented credential to its issuance commitment exactly. The XOF, rejection-sampling order, byte encoding, and maximum iteration policy are part of the protocol specification so that seed expansion is deterministic and cross-platform reproducible.

Thus issuance has the two-message form

$$\text{User} \longrightarrow \text{Signer} : C, \{z_{i,\chi_i}, \mathbf{c}_{i,1-\chi_i}\}_i, \quad \text{Signer} \longrightarrow \text{User} : h, \pi_{\text{resp}}.$$

Fiat–Shamir removes the challenge round but permits offline transcript grinding. Prescribing k challenge bits costs about 2^k classical random-oracle queries and about $2^{k/2}$ generic quantum queries. A tight QROM proof is therefore required for the post-quantum parameter choice. A fresh signer nonce limits precomputation, but adds a message unless it is already supplied by an authenticated session setup.

3.3 Aggregate response

The signer derives a bounded aggregate error deterministically from the entire selected transcript,

$$e = H_{D_e}(K, \text{sid}, \mathbf{c}_0^*, \dots, \mathbf{c}_{\kappa-1}^*),$$

and computes only

$$h = \sum_{i=0}^{\kappa-1} \mathbf{s}_i^T \mathbf{c}_i^* + e.$$

No individual lane value is released or made observable through timing or errors. The signer proves the batched relation

$$\mathcal{R}_{\text{resp}} : \begin{cases} h = \sum_i \mathbf{s}_i^T \mathbf{c}_i^* + e, \\ \mathbf{t}_i^T = \mathbf{s}_i^T B_i + \mathbf{e}_{1,i}^T \\ \|\mathbf{s}_i\| \leq \beta_s, \|\mathbf{e}_{1,i}\| \leq \beta_e, \|e\| \leq \beta_e. \end{cases} \quad \text{for every } i,$$

The witness is the tuple of all component secrets and errors. Proving knowledge only of $S = \sum_i \mathbf{s}_i$ is insufficient because the $\mathbf{c}_i^*$ differ. The important efficiency observation is that $\mathcal{R}_{\text{resp}}$ consists only of lattice-linear relations and norm bounds; it contains no hash evaluation.

3.4 Unblinding and presentation

After verifying the response proof, the user computes

$$v = \left[h - \sum_i \mathbf{t}_i^T \mathbf{r}_i^* \right]_p$$

and stores

$$\sigma = ((m_i^*, \rho_i^*)_{i=0}^{\kappa-1}, v), \quad (m_i^*, \rho_i^*) = (m_{i,b_i^*}, \rho_{i,b_i^*}).$$

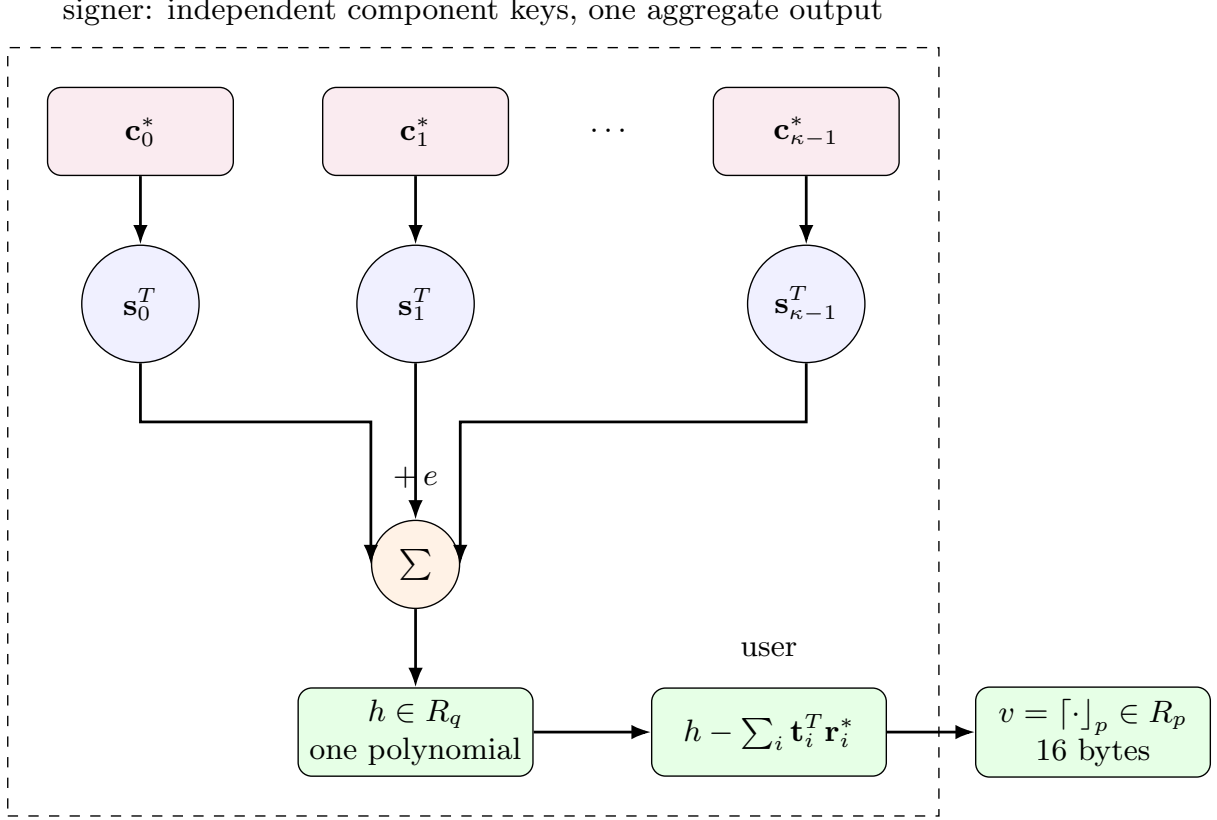


Figure 2: Aggregate-key evaluation. Each selected commitment is evaluated under an independent component key, but individual lane responses are never released. Summation collapses the vector of evaluations to one polynomial; the user then unblinds and rounds it to the compact value v .

At presentation, the keyed verifier accepts iff

$$v = \left\lfloor \sum_i s_i^T H_R(\text{credential-lane}, i, m_i^*, \rho_i^*) \right\rfloor_p.$$

The issuance session identifier must not enter the presented lane hash: doing so would give the issuer a trivial linkage handle. Session binding belongs in the cut-and-choose commitments and challenge, while issuance and presentation use the same credential-domain lane hash. An application-level payload can be defined as

$$M = H_{\text{final}}(\text{issuer}, \text{type}, (m_i^*, \rho_i^*)_{i=0}^{\kappa-1}).$$

4 Correctness

For an honestly generated selected branch,

$$\mathbf{c}_i^* = B_i \mathbf{r}_i^* + \mathbf{e}_{2,i}^* + \mathbf{u}_i^*.$$

Consequently,

$$h - \sum_i \mathbf{t}_i^T \mathbf{r}_i^* = \sum_i \mathbf{s}_i^T \mathbf{u}_i^* + \sum_i (\mathbf{s}_i^T \mathbf{e}_{2,i}^* - \mathbf{e}_{1,i}^T \mathbf{r}_i^*) + e.$$

Thus issuance and verification agree whenever the aggregate error does not cross a rounding boundary. Compared with BLNS23, the error contains a sum of κ lane errors. Parameters therefore cannot be copied unchanged: the noise bound grows at least as $O(\sqrt{\kappa})$ for independent errors and as $O(\kappa)$ under a worst-case norm bound. This may require a larger q , a smaller rounding modulus p , or narrower lane distributions.

5 Rectangle attacks and lane freshness

The coordinate-wise aggregate MAC is additively separable. If a valid lane input can be reused across issuance sessions, this gives a one-more forgery even when every commitment is honestly formed. Ignore the small correctness error momentarily and define

$$F_i(x) = \mathbf{s}_i^T H_R(x), \quad Y(x_0, \dots, x_{\kappa-1}) = \sum_i F_i(x_i).$$

Here $x_i = (m_i, \rho_i)$ and Y is the value available to the user before final rounding. Fix every coordinate except two, denoted a and b , and choose two distinct valid inputs x_a^0, x_a^1 and x_b^0, x_b^1 . Let Y_{uv} denote the aggregate with $x_a = x_a^u$ and $x_b = x_b^v$. Three honest issuance queries give

$$\begin{aligned} Y_{00} &= F_a(x_a^0) + F_b(x_b^0) + K, \\ Y_{10} &= F_a(x_a^1) + F_b(x_b^0) + K, \\ Y_{01} &= F_a(x_a^0) + F_b(x_b^1) + K, \end{aligned}$$

where K is the fixed contribution of every other lane. The user computes

$$\begin{aligned} Y^* &= Y_{10} + Y_{01} - Y_{00} \\ &= F_a(x_a^1) + F_b(x_b^1) + K = Y_{11}. \end{aligned}$$

Thus three issued vector credentials yield the fourth corner, which was never issued. All four message vectors are pairwise distinct.

In the real protocol, the user has unblinded values

$$\tilde{Y}_{uv} = h_{uv} - \sum_i \mathbf{t}_i^T \mathbf{r}_{i,uv} = Y_{uv} + \eta_{uv},$$

where each η_{uv} is a small correctness error. It computes

$$\tilde{Y}^* = \tilde{Y}_{10} + \tilde{Y}_{01} - \tilde{Y}_{00} = Y_{11} + (\eta_{10} + \eta_{01} - \eta_{00})$$

and rounds $\tilde{Y}^*$. The combined error grows by only a constant factor. With the correctness margin required by the scheme, it crosses a rounding boundary only with negligible probability. The resulting MAC for the unqueried vector (x_a^1, x_b^1) therefore verifies.

Theorem 1 (Three-query forgery under reusable lane inputs). *For $\kappa \geq 2$, suppose valid values $x_i = (m_i, \rho_i)$ may be repeated in different issuance sessions and the correctness error remains negligible under a three-term signed sum. Then the coordinate-wise aggregate construction is not one-more unforgeable: three honest issuance queries produce four pairwise-distinct valid vector-message credentials.*

The same identity refutes any stronger aggregate-HMLWE assumption that permits cross-query reuse of valid coordinate inputs. Four oracle queries satisfy

$$\tilde{Y}_{00} + \tilde{Y}_{11} - \tilde{Y}_{10} - \tilde{Y}_{01} = \eta_{00} + \eta_{11} - \eta_{10} - \eta_{01},$$

which is small. For a true random oracle, the left-hand side is uniform in R_q . A centered-norm test therefore distinguishes the aggregate oracle from random with overwhelming advantage.

In the current construction, however,

$$\rho_{i,b} = H_\rho(\text{sid}, i, b, \mathbf{r}_{i,b}, \mathbf{e}_{2,i,b})$$

contains a signer-backed, globally unique sid . Reusing the same $x_i = (m_i, \rho_i)$ in a second session would require finding a cross-session preimage or collision for H_ρ . The rectangle attack therefore motivates a mandatory rule: session identifiers must never repeat within a key epoch, and every valid lane input used in the aggregate-HMLWE game must be fresh across issuance queries. The fully honest attack does not prove that a freshness-restricted game is secure; it establishes that no theorem without the restriction can be true. A repaired proof must also cover concurrency, rollback, and distributed replay-state failures that could violate uniqueness.

Session freshness does not remove every use of the rectangle identity. A selected malformed branch need not claim the old message: because it is never opened, its residual can simply copy a previously observed hash coefficient. Partition the lanes into $A \dot{\cup} B = [\kappa]$. First obtain an honest baseline

$$Y_{00} = \sum_{i \in A} F_i(x_i^0) + \sum_{i \in B} F_i(x_i^0).$$

In a second issuance use fresh honest inputs x_i^1 in A and copied, malformed baseline coefficients in B ; in a third do the converse. Conditional on the required Fiat–Shamir challenges, the resulting values are

$$\begin{aligned} Y_{10} &= \sum_{i \in A} F_i(x_i^1) + \sum_{i \in B} F_i(x_i^0), \\ Y_{01} &= \sum_{i \in A} F_i(x_i^0) + \sum_{i \in B} F_i(x_i^1). \end{aligned}$$

Hence $Y_{10} + Y_{01} - Y_{00} = Y_{11}$ as before, although every *valid* input is session-fresh. The second issuance requires selecting $|B|$ malformed branches and the third requires selecting $|A|$. A balanced partition therefore costs about $2^{\kappa/2}$ classical random-oracle trials, or $2^{\kappa/4}$ generic quantum trials. This is still exponential, but it shows that analyzing only the all- κ -malformed event overstates concrete security by a square in the classical ROM and by a fourth-power in the QROM.

The balanced attack is optimal within a natural copy-and-combine class. Model each distinct lane coefficient as an independent formal atom and call an occurrence a *copy* if the same atom occurred earlier in that lane.

Theorem 2 (Half-copy lower bound for formal linear splicing). *Let $Q \geq 2$ issued aggregates be combined with nonzero coefficients $\alpha_1, \dots, \alpha_Q \in R_q$ to produce an unissued aggregate containing one target atom in every lane. If the identity holds for every assignment of the lane atoms, then some issuance after the first contains copies in at least $\kappa/2$ lanes.*

Proof. Fix a lane and group the Q occurrences according to equal atoms. The group containing the target atom must have coefficient sum one; every other group must have coefficient sum zero. Because every α_j is nonzero, a zero-sum group has at least two occurrences. At most one group—the target group—can be a singleton. Consequently at most $1 + \lfloor (Q-1)/2 \rfloor$ different atoms occur in this lane, so at least $\lceil (Q-1)/2 \rceil$ occurrences are copies. Summing over all κ lanes gives at least $\kappa \lceil (Q-1)/2 \rceil$ copy incidences. No occurrence in the first issuance is a copy. Averaging over the remaining $Q-1$ issuances therefore puts at least

$$\frac{\kappa \lceil (Q-1)/2 \rceil}{Q-1} \geq \frac{\kappa}{2}$$

copies in one issuance. □

With session-bound validity, such copied atoms must occur in unopened malformed branches except with a hash collision. Thus every formal linear splice fixes at least $\kappa/2$ challenge bits in one issuance. This theorem does not rule out lattice-specific attacks using short nontrivial relations among distinct hash vectors; excluding those is part of the mfr-AHMLWE assumption.

6 Majority-fresh aggregate HMLWE

We isolate the exact pseudorandomness property needed after the usual BLNS23 hybrids have removed the public blinding terms. This abstraction intentionally separates the lattice question from Fiat–Shamir extraction.

Let $G_i : \{0, 1\}^* \rightarrow R_q^n$ be independent random oracles, let $\mathbf{s}_i \leftarrow \$ D_s^n$ for $i \in [\kappa]$, and let D_E be the aggregate error distribution. In query j , an adaptive adversary submits

$$(\mathbf{a}_{j,0}, \dots, \mathbf{a}_{j,\kappa-1}), \quad V_j \subseteq [\kappa], \quad |V_j| > \kappa/2,$$

together with strings $\mu_{j,i}$ for $i \in V_j$. The query is admissible if

$$\mathbf{a}_{j,i} = G_i(\mu_{j,i}) \quad (i \in V_j)$$

and every pair $(i, \mu_{j,i})$ is fresh among earlier admissible queries. The coordinates outside V_j are arbitrary and may depend on the entire previous transcript and on all random-oracle answers seen so far. The real oracle returns

$$z_j = \sum_{i=0}^{\kappa-1} \mathbf{s}_i^T \mathbf{a}_{j,i} + e_j, \quad e_j \leftarrow \$ D_E,$$

whereas the ideal oracle returns an independent uniform element of R_q . The *majority-fresh aggregate HMLWE* (mfr-AHMLWE) assumption states that these two adaptive experiments are computationally indistinguishable. In the protocol, $\mu_{j,i}$ represents the session-bound pair $(m_{j,i}, \rho_{j,i})$ and V_j is the set of selected branches for which the extractor obtains valid witnesses. A complete reduction must additionally show that the real aggregate error is distributed as, or indistinguishable from, D_E .

The majority and freshness conditions are stronger than mere uniqueness of the complete message vector: it forbids reuse of a valid coordinate under the same lane key and requires fewer than half the selected coordinates to be malformed. Section 5 proves both restrictions are necessary for the claimed concrete security. The definition does not restrict an arbitrary malformed coordinate from repeating a previous ring element; that behavior is deliberately included in the assumption.

Lemma 1 (Fixed honest lane special case). *Suppose there is an index i^* such that $i^* \in V_j$ in every aggregate query. Then mfr-AHMLWE reduces to ordinary HMLWE for lane i^* , with a factor at most κ if the reduction must guess i^* .*

Proof. Embed the HMLWE secret in lane i^* and sample every other lane secret locally. On query j , freshness gives a new HMLWE message μ_{j,i^*} . Obtain $b_j = \mathbf{s}_{i^*}^T G_{i^*}(\mu_{j,i^*}) + e_j$ from the HMLWE oracle and return

$$b_j + \sum_{i \neq i^*} \mathbf{s}_i^T \mathbf{a}_{j,i}.$$

This is a perfect real simulation. If b_j is uniform, adding the locally computed term preserves uniformity and gives the ideal experiment. The arbitrary coordinates cause no simulation problem because their corresponding secrets were sampled by the reduction. $\square$

Corollary 1 (All lanes valid). *If $V_j = [\kappa]$ in every query, mfr-AHMLWE reduces to ordinary HMLWE (and hence, in the random-oracle model, to MLWE by BLNS23 Lemma 4.3).*

Proof. Choose any fixed lane i^* and apply the preceding lemma. This route is preferable to treating the concatenation of all lane hashes as one atomic random oracle: the adversary may query the component random oracles separately before deciding which values to combine. $\square$

Neither lemma proves the general game. Majority sets intersect pairwise, but polynomially many majority sets need not have a common element. Thus one may still have no lane valid in every query. A reduction that embeds its unknown HMLWE secret in lane i must then answer queries in which $i \notin V_j$ and $\mathbf{a}_{j,i}$ is adversarially chosen. Doing so requires the unavailable value $\mathbf{s}_i^T \mathbf{a}_{j,i}$; an ordinary HMLWE oracle only evaluates fresh hashed inputs. Guessing the lane or signing query does not supply these missing evaluations. Thus the standard BLNS23 reduction proves the two lemmas above but not the adaptive rotating-cover case. We make the rotating-majority case an explicit assumption rather than silently treating it as a consequence of MLWE.

Theorem 3 (Conditional adaptive aggregate masking). *Assume binding of the global commitment, simultaneous extraction of every valid selected branch needed to identify V_j , the BLNS23 hybrids mapping valid lane residuals to $G_i(\mu)$ with aggregate error D_E , and mfr-AHMLWE. For a classical adversary making at most q_H Fiat–Shamir trials, all accepted aggregate responses may be replaced by independent uniform ring elements with distinguishing loss at most*

$$\text{Adv}_{\text{mfr-AHMLWE}} + q_H 2^{-\kappa/2} + \varepsilon_{\text{ext}} + \varepsilon_{\text{hyb}},$$

where ε_{ext} and ε_{hyb} are the extraction and BLNS23-hybrid failure probabilities. In the QROM, the generic challenge-search term becomes $O(q_H^2 2^{-\kappa/2})$, in addition to the losses of the required QROM extractor.

Proof. Except with extraction failure, let V_j contain all valid selected branches. If $|V_j| \leq \kappa/2$, at least $\kappa/2$ selected branches are malformed. For a fixed committed branch vector, acceptance then fixes at least $\kappa/2$ independent challenge bits. A union bound over classical Fiat–Shamir trials gives $q_H 2^{-\kappa/2}$; generic quantum search gives the stated quadratic-query bound. Conditioned on the complementary event, every

accepted query satisfies $|V_j| > \kappa/2$. Session-bound ρ values make its valid lane messages fresh except with a hash collision. After the stipulated BLNS23 hybrids, the simulator submits the extracted V_j , valid messages, and arbitrary remaining residuals to its mfr-AHMLWE oracle. The real oracle reproduces the aggregate response; the ideal oracle supplies the uniform replacement. Adding the bad-event and hybrid losses proves the claim. $\square$

7 Cut-and-choose analysis

Call a branch *openable* if its disclosed witness passes every opening check. An accepted transcript has at least one openable branch in every pair. Suppose the adversary prepares one openable and one malformed branch in each of k attacked lanes. All malformed branches are selected exactly when the challenge opens the openable branch in every attacked lane, hence

$$\Pr[\text{all attacked selections malformed}] = 2^{-k}.$$

In particular, removing every honest mask across κ lanes has probability $2^{-\kappa}$ per attempt, and at most $Q2^{-\kappa}$ over Q attempts by a union bound.

For the direct scaled-unit attack, the adversary sets every malformed selected commitment to $\alpha \mathbf{e}_j$. If all lanes select those branches, then

$$h = \alpha \sum_i s_{i,j} + e,$$

so rounding h/α exposes a coordinate of the summed key. The event has probability $2^{-\kappa}$ per independent classical trial, while generic quantum Fiat–Shamir grinding costs about $2^{\kappa/2}$. If any unopened lane is honest, its term remains in the aggregate and is intended to computationally mask the malformed terms.

This last sentence is an assumption to be proved, not a consequence of the elementary probability calculation.

7.1 Extraction needed by the majority-fresh game

The desired statement can be separated into an information-selection claim and a lattice-pseudorandomness claim. Let $\text{Valid}(i, b)$ denote the existence of an opening satisfying all lane relations. For a fixed vector of branch commitments, there is at most one challenge χ for which every opened branch is valid while every selected branch is invalid: in every pair it must open the unique valid branch. Consequently, a classical adversary making q_H Fiat–Shamir trials reaches this all-invalid event with probability at most

$$q_H 2^{-\kappa},$$

up to collisions and challenge bias. In the QROM the corresponding generic bound has the expected quadratic loss, $O(q_H^2 2^{-\kappa})$.

For one isolated issuance, knowledge of one valid selected witness is enough to invoke an HMLWE mask after sampling every other lane key locally. Across adaptive issuances this argument is sound only if the same embedded lane stays valid, as proved in Section 6. It does not establish the rotating-majority game: when the embedded lane is malformed, the simulator is asked for an arbitrary inner product under an unknown secret.

The classical extraction step has a direct two-fork argument. Fix the user’s state immediately before its Fiat–Shamir query, and obtain two accepting transcripts with the same branch commitments but distinct challenges $\chi \neq \chi'$. There is an index j such that $\chi_j \neq \chi'_j$. Because the challenges are binary,

$$\chi'_j = 1 - \chi_j = b_j^*.$$

The second transcript therefore opens exactly the branch that was selected and left unopened in the first transcript. Verification of that opening yields $(m_j^*, \rho_j^*, \mathbf{r}_j^*, \mathbf{e}_{2,j}^*)$, the witness required by the HMLWE reduction.

Lemma 2 (Two-fork selected-lane extraction). *Assuming binding of the global branch commitment, two accepting transcripts with identical pair commitments and distinct challenge vectors reveal a valid witness for at least one branch selected in the first transcript (and symmetrically for one branch selected in the second).*

The proof is the coordinate argument above and is information theoretic, but it extracts only one selected branch. The conditional theorem in Section 6 needs the witnesses of a strict majority of selected branches. Repeating the fork does not automatically provide them: accepting challenges can differ repeatedly in the same small set of coordinates. A multi-fork extractor must either force sufficiently broad challenge coverage or use an independently extractable branch commitment.

If a fixed commitment vector admits only one accepting challenge, its Fiat–Shamir success probability is $2^{-\kappa}$. If a classical polynomial-time prover succeeds with non-negligible probability, the standard forking strategy—rewind at the challenge query and answer it independently—obtains two distinct accepting challenges with non-negligible probability, subject to the usual forking loss in the number of random-oracle queries. Combining this extractor with the preceding HMLWE hybrid gives a plausible classical-ROM proof route.

What remains unresolved is making the extractor online and adaptive for polynomially many signing queries and concurrent sessions, and then lifting it to the QROM. Generic Fiat–Shamir proofs of knowledge do not automatically provide adaptive multi-theorem extraction, and quantum rewinding cannot simply copy the prover state. A QROM proof would instead need a measure-and-reprogram argument and a suitable adaptive special-soundness theorem [4].

One clean but expensive proof device is to add an extractable commitment to each branch witness, or attach an encryption-to-the-sky of each branch witness. The security reduction then knows the extraction key and can recover the selected valid witness, while no such key exists in the real protocol. This avoids adaptive rewinding but adds 2κ large lattice ciphertexts and may erase the proposed efficiency advantage. Finding a lightweight extractability mechanism is therefore part of the contribution.

7.2 Counterexamples to nearby variants

The analysis already gives concrete counterexamples to several tempting simplifications:

1. If individual lane responses are released, attacking one pair succeeds with probability $1/2$ and leakage accumulates across sessions.
2. If all lanes reuse the same secret $\mathbf{s}$, a malformed lane can be correlated with an honest lane so that their coefficients cancel. Independent lane keys are essential.
3. If the Fiat–Shamir hash omits any branch commitment, public key, lane index, or protocol domain, the omitted value can be chosen after the challenge.
4. If $\kappa = 128$ is used against quantum users, the balanced splice has only approximately 32 bits of generic quantum work.
5. If responses or partial sums are emitted before all openings pass, one malformed lane again becomes a probability- $1/2$ key-leakage oracle.

The unrestricted “one-honest-lane aggregate security” conjecture, which allows reuse of a valid lane input, is false by Section 5. For globally fresh valid inputs, Section 6 proves the all-lanes-valid and fixed-honest-lane cases from ordinary HMLWE. Security with an adaptively rotating majority of valid lanes is exactly the remaining mfr-AHMLWE assumption; no reduction to ordinary MLWE is presently claimed.

8 Blindness intuition

The issuer learns only the opened decoy branch in each lane. The credential contains the complementary messages. At presentation the issuer learns $(m_i^*, \rho_i^*)_i$ and can subtract their hashes from the stored unopened commitments, but deciding whether

$$\mathbf{c}_i^* - H_R(m_i^*, \rho_i^*) = B_i \mathbf{r}_i^* + \mathbf{e}_{2,i}^*$$

for short unknown values should be hidden by the underlying MLWE distribution. This suggests issuance–presentation unlinkability when opened and unopened messages have indistinguishable distributions.

The response proof is necessary against a malicious issuer. Without it, the issuer may use malformed component keys or signer-chosen errors to watermark credentials. The proof must establish key-parameter consistency for every lane and at least constrain the aggregate error norm. Proving deterministic

PRF generation of the error would add a nonlinear relation and may undermine the claimed efficiency; alternatively, one must prove that every permitted bounded error is erased by rounding except with negligible probability and cannot be used for selective-failure tagging.

9 Requirements for a full proof

1. **Fresh aggregate pseudorandomness.** Prove mfr-AHMLWE from a standard lattice assumption, or state and cryptanalyze it as a new assumption. The proof must permit arbitrary malformed coordinates and an adaptively rotating strict majority of fresh valid coordinates.
2. **Global lane freshness.** Enforce non-repeating session identifiers across concurrency, crashes, rollback, and distributed signer replicas, and bind ρ to that identifier as in the construction above.
3. **Related inputs across sessions.** Prove security for fresh but algebraically related message vectors; coordinate freshness blocks the fully honest rectangle but not the balanced malformed-copy splice, and does not by itself establish pseudorandomness.
4. **Selective failure.** Show that malicious challenges, aborts, malformed response proofs, and rounding failures do not identify the hidden branch vector at later presentation.
5. **Concurrency and reset security.** Bind every message to a unique session and analyze interleaved, replayed, reset, and rolled-back sessions.
6. **One-more unforgeability.** Define the repaired message space and rule out linear combinations or splicing of issued credentials.
7. **Malicious-key security.** Prove that the batched response NIZK gives key-parameter consistency and prevents subliminal tagging.
8. **Noise growth.** Instantiate parameters for the aggregate error and bound the rounding-failure probability at the target security level.
9. **Fiat-Shamir and challenge entropy.** Give ROM and QROM analyses accounting for adaptive transcript grinding, multi-target attacks, malicious abort-and-restart strategies, and concurrency. Determine whether $\kappa = 512$ tightly achieves 128-bit post-quantum soundness after accounting for the balanced splicing attack.
10. **Side channels and atomicity.** Ensure that no individual lane response, partial sum, failing index, or timing signal is released.

10 Preliminary efficiency discussion

The candidate removes the hash-heavy user NIZK, replacing it with 2κ hash commitments and κ seeded openings. For a classical target with $\kappa = 256$, issuance prepares 512 lattice commitments and opens 256 witnesses; the post-quantum starting point $\kappa = 512$ doubles those figures. Fiat-Shamir preserves two-message issuance despite this bandwidth. The signer performs κ hash-and-relation checks and one aggregate evaluation. The response NIZK is larger than the single-lane BLNS23 π_2 , but is batchable and purely lattice algebraic.

10.1 Concrete communication estimate

We instantiate the wire calculation with the BLNS23 keyed-verification values $d = 64$, $n = 93$, $\lceil \log_2 q \rceil = 152$, $p = 4$, and 32-byte messages and tags. We take $\kappa = 512$ as the minimum starting point for 128-bit post-quantum resistance to the balanced Fiat-Shamir splice. Ring elements are packed at 152 bits per coefficient. Branch openings are 32-byte seeds and the single global branch commitment is a 48-byte digest.

One commitment vector has exact packed length

$$|c| = nd \lceil \log_2 q \rceil = 93 \cdot 64 \cdot 152 = 904,704 \text{ bits} = 113,088 \text{ bytes}.$$

Only the $\kappa = 512$ unopened vectors are transmitted in full, occupying 57,901,056 bytes. The global commitment occupies 48 bytes and the 512 opened seeds occupy 16,384 bytes. With a 32-byte session identifier, the exact fixed portion of the first message is therefore 57,917,520 bytes (55.23 MiB). The challenge is recomputed and is not transmitted separately. Relative to transmitting all commitment vectors and explicit witnesses, seeded openings reduce the first message by approximately 52.5%.

The aggregate response $h \in R_q$ occupies

$$|h| = d \lceil \log_2 q \rceil = 64 \cdot 152 = 9,728 \text{ bits} = 1,216 \text{ bytes}.$$

The second message has length $1,216 + |\pi_{\text{resp}}|$ bytes. An exact total for that message requires an instantiated batched LaBRADOR relation and cannot be inferred from the asymptotic construction.

Wire component	Bytes
512 unopened commitment vectors	57,901,056
One 48-byte global commitment	48
512 32-byte opened seeds	16,384
Session identifier	32
User $\rightarrow$ signer, fixed total	57,917,520
Aggregate h	1,216
Signer $\rightarrow$ user	$1,216 + \pi_{\text{resp}} $
MAC $(\rho_0, \dots, \rho_{511}, v)$	16,400
MAC plus 512 32-byte messages	32,784

Table 1: Preliminary exact wire sizes with seeded openings and one 384-bit global branch commitment under the stated canonical encoding. The batched response proof is not yet instantiated.

The MAC calculation uses 512 tags of 32 bytes and $d \log_2 p = 128$ bits for v :

$$|\text{MAC}| = 512 \cdot 32 + 16 = 16,400 \text{ bytes}.$$

If the messages are considered part of the stored credential rather than external signed data, 16,384 additional bytes give a 32,784-byte credential.

10.2 Optional optimization: higher-arity cut-and-choose

The binary construction above is the base scheme: it gives the simplest protocol description, two-fork extractor, and security game. Higher arity is an optional optimization and should be analyzed only after the binary scheme is proved secure.

The dominant term can be reduced by replacing binary pairs with groups of $A = 2^a$ branches. Fiat–Shamir selects one branch per group to remain unopened; the other $A - 1$ branches are disclosed as 32-byte seeds. If there are g independent groups, making every selected branch malformed succeeds with probability

$$A^{-g} = 2^{-ag}.$$

The balanced splice constrains about half of the groups, so its generic quantum cost is approximately $2^{ag/4}$. For a 128-bit post-quantum target we therefore require $ag \geq 512$. Only g full lattice vectors are then transmitted. Ignoring the response proof, the fixed first-message size is

$$g|\mathbf{c}| + g(A - 1) \cdot 32 + 48 + 32 \text{ bytes}, \quad g = \left\lceil \frac{512}{\log_2 A} \right\rceil.$$

For bandwidth alone, the stated parameters are minimized near $A = 512$, where the first message is 7,378,160 bytes (7.04 MiB): 57 full vectors consume 6,446,016 bytes, 29,127 opened seeds consume 932,064 bytes, and the global commitment plus session identifier consume 80 bytes. The corresponding MAC contains 57 tags and one 16-byte rounded value, hence

$$57 \cdot 32 + 16 = 1,840 \text{ bytes}.$$

This setting requires generating and checking 29,184 branches, however, versus 1,024 in the binary construction. Values such as $A = 16$ or $A = 32$ may therefore give a better end-to-end tradeoff:

A	g	challenge bits	generated branches	first message
2	512	512	1,024	55.23 MiB
4	256	512	1,024	27.63 MiB
16	128	512	2,048	13.86 MiB
32	103	515	3,296	11.21 MiB
64	86	516	5,504	9.44 MiB
256	64	512	16,384	7.40 MiB
512	57	513	29,184	7.04 MiB
1024	52	520	53,248	7.23 MiB

Table 2: Bandwidth/computation tradeoff for higher-arity cut-and-choose. The branch count approximates user generation and signer opening work.

they reduce issuance to 13.86 or 11.21 MiB while increasing branch computation by only two to three times. A benchmark must include deterministic expansion, hash-to-ring, matrix-vector products, and response-proof scaling before selecting A .

The two-fork extraction argument generalizes: two accepting transcripts whose selected indices differ in some group cause the second transcript to open the branch selected in the first. The proof must use an A -ary challenge and bind all branches in the global commitment.

This optimization changes the key shape and is not merely a wire encoding. In the binary base scheme there are $\kappa = 512$ independent component secrets

$$(\mathbf{s}_0, \dots, \mathbf{s}_{511}) \in (R_q^n)^{512}.$$

In the A -ary scheme there are only $g = \lceil 512 / \log_2 A \rceil$ component secrets,

$$(\mathbf{s}_0, \dots, \mathbf{s}_{g-1}) \in (R_q^n)^g, \quad h = \sum_{i=0}^{g-1} \mathbf{s}_i^T \mathbf{c}_i^* + e.$$

Thus arity changes the public and secret key sizes, the number of terms and noise contributions in the aggregate, the batched response-proof statement, and the number of (m_i, ρ_i) pairs in the MAC. Each component key retains the full BLNS23 module dimension; only the number of independently keyed groups changes. A concrete A must therefore receive its own correctness, hardness, and parameter analysis rather than inheriting the binary parameters automatically.

The credential contains κ message/tag pairs unless they admit a safe compression mechanism. This is a major cost and must be included in any comparison. A credible evaluation should report total issuance bytes, prover and verifier time, credential bytes, presentation time, public-key size, secret key size, and rounding-failure probability.

11 Conclusion

The cut-and-choose mechanism detects malformed selections, but analyzing only the all-malformed event is insufficient. Reusable valid inputs give a free four-corner forgery, while session freshness still permits a balanced malformed-copy splice costing about $2^{\kappa/2}$ classically or $2^{\kappa/4}$ under generic quantum search. The formal half-copy theorem shows this cost is optimal for copy-and-combine linear splicing, but it does not exclude lattice-specific relations. Ordinary HMLWE proves aggregate masking when all lanes are valid or one fixed lane remains valid throughout; it does not cover an adaptively rotating valid majority. The mfr-AHMLWE assumption, simultaneous majority extraction, QROM soundness, malicious-key security, and concrete aggregate-noise parameters therefore remain necessary before the proposal can support a security claim.

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